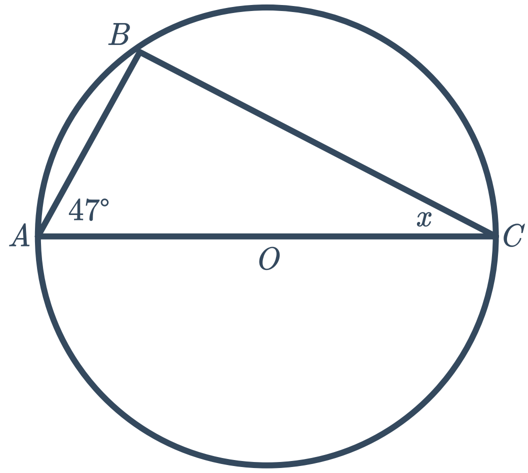
Missing angles

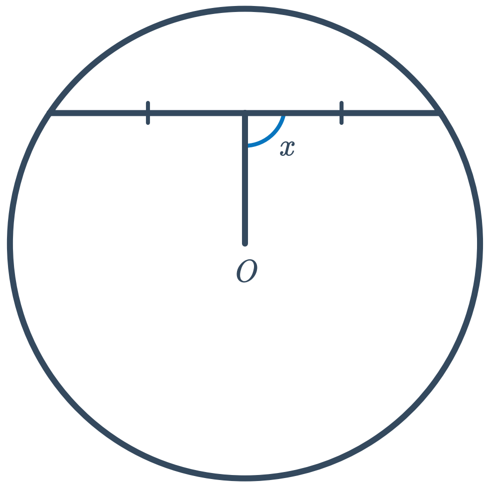
1 In the given diagram, AC is a straight line passing through centre O:

Solve for x.



2 Consider the circle with centre O shown:

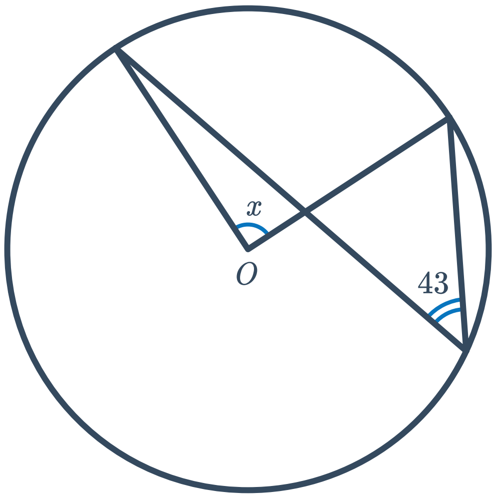
Solve for x.



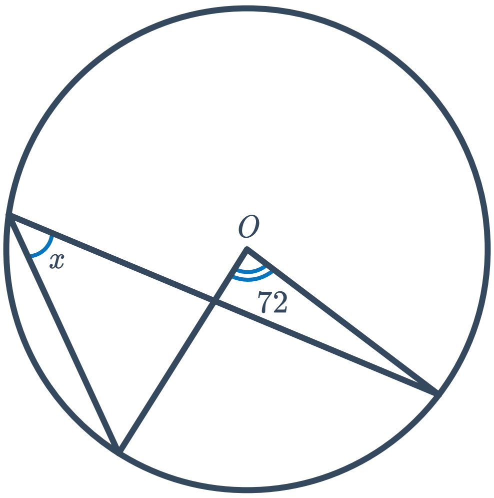
Missing angles (Extended)

3 For each of the following circles with centre O, solve for x:

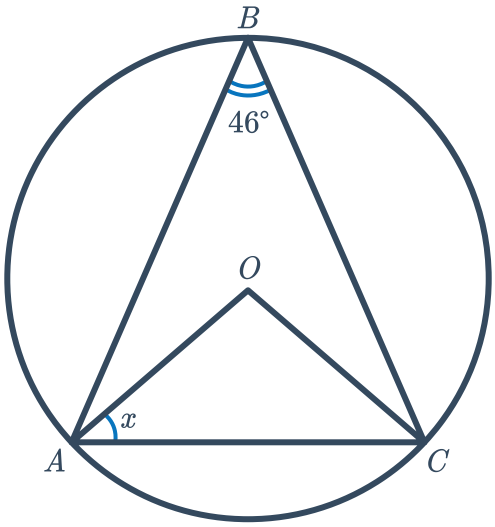
a



b

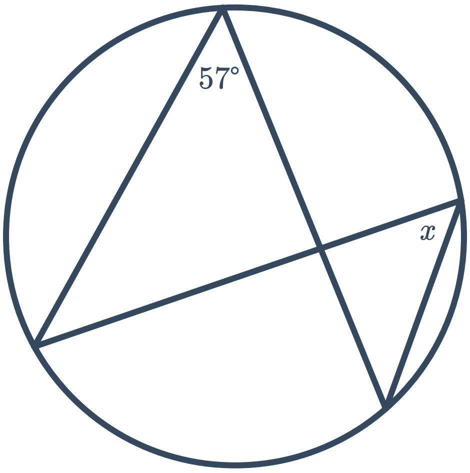


c

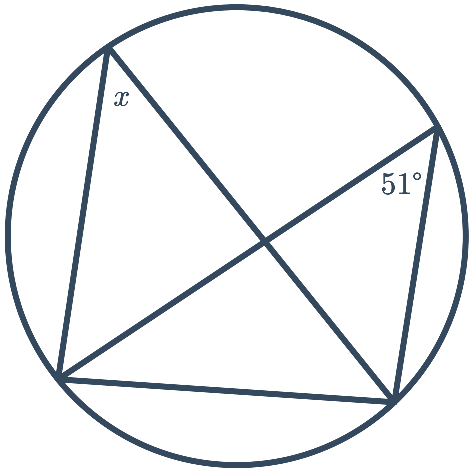


4 For each of the following circles, solve for [redacted]

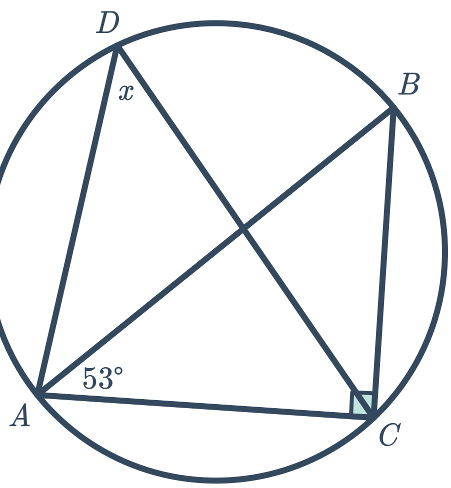
a



b

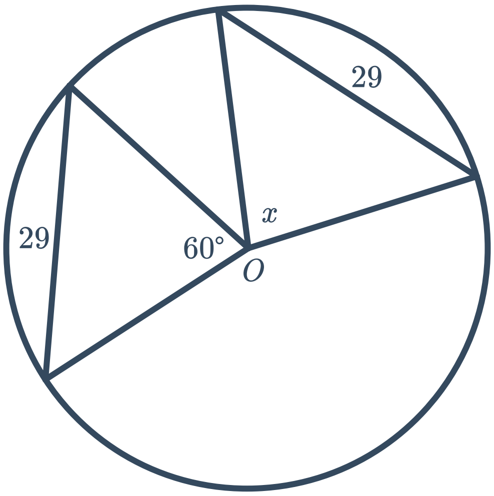


c

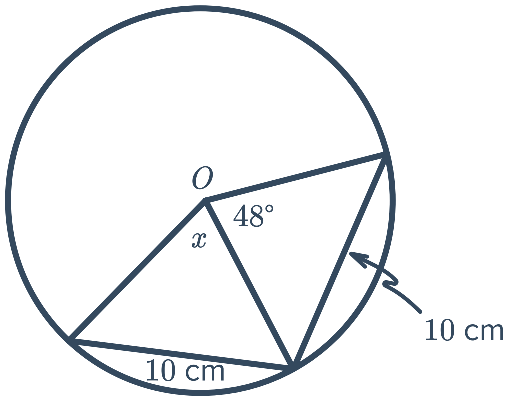


5 For each of the following circles with cent[redacted] solve for x:

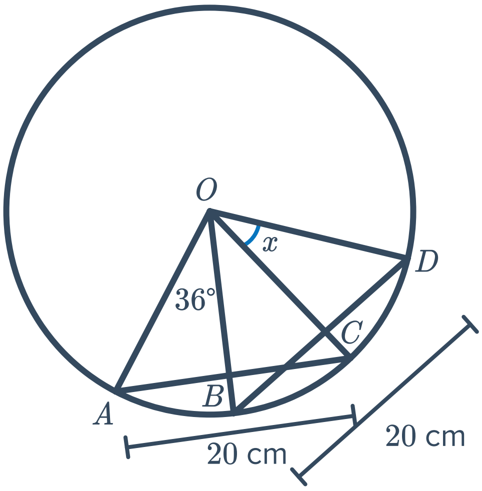
a



b



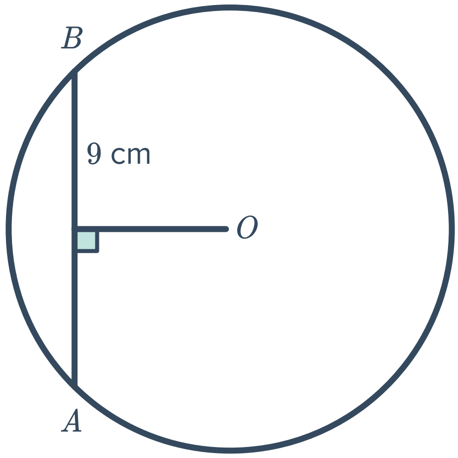
c



Missing lengths

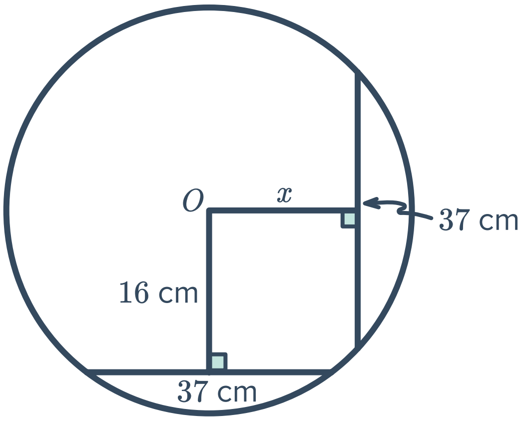
6 Consider the following circle where O is the centre:

Find the length AB.

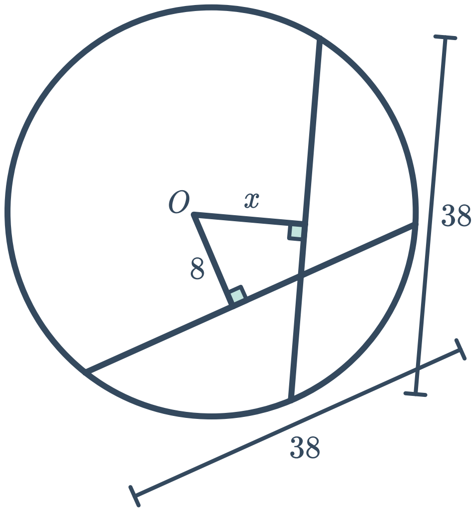


7 For each of the following circles with centre O, solve for x:

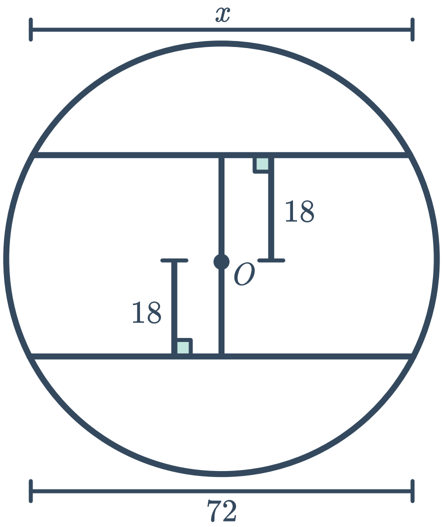
a



b

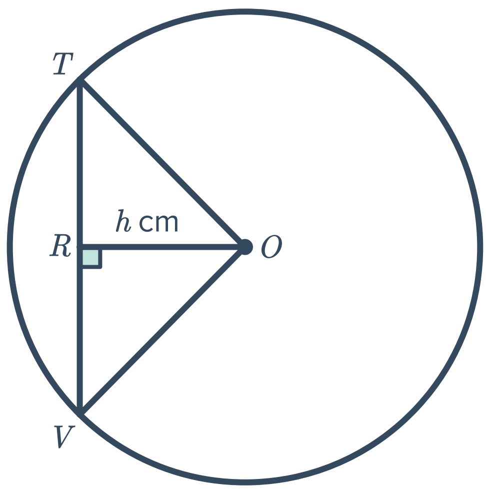


c

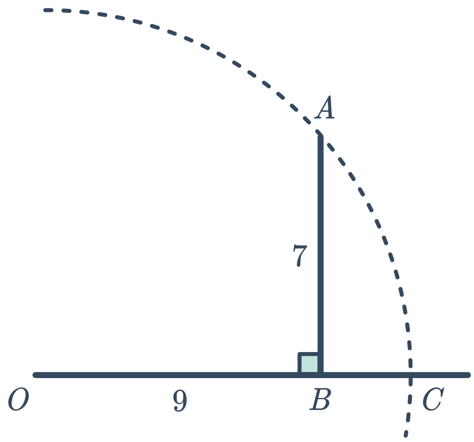


8 The given circle has $TR = 12$ cm, the radius is 13 cm, and OR has length h cm.

Find the area of $\triangle TOV$.



9 The diagram shows AC as the arc of a circ
with O as its centre, $OB = 9$, and $AB = 7$:



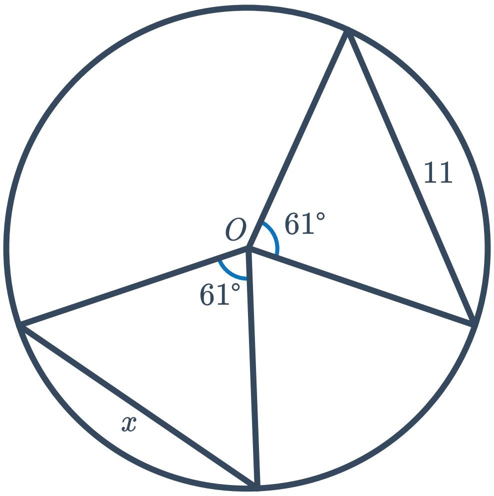
a Find the exact length of OA.

b Hence, find the exact length of BC.

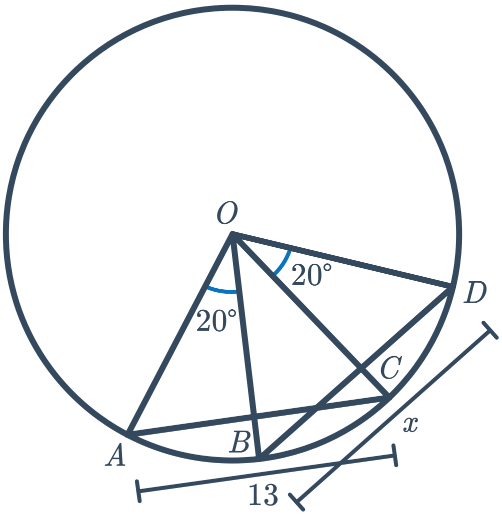
Missing lengths (Extended)

10 For each of the following circles with centre O, solve for x:

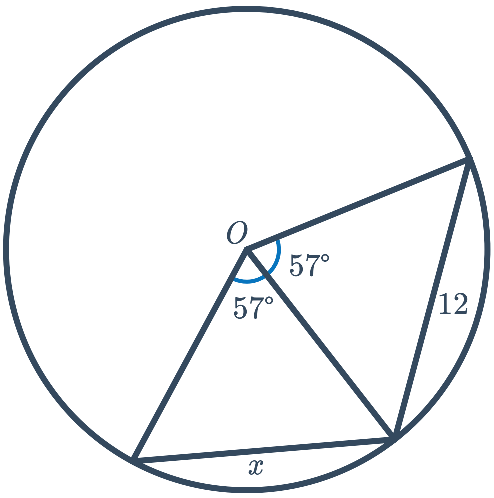
a



b



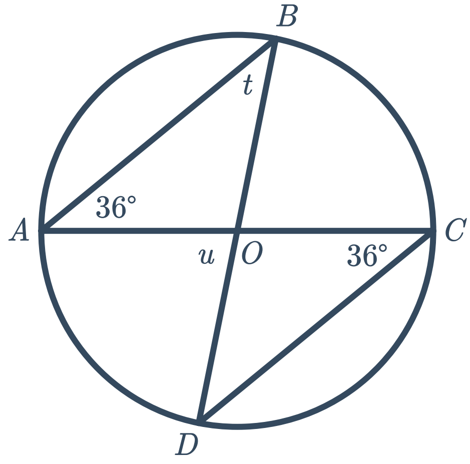
c



11 Consider the following circle where O is the centre:

a Solve for t.

b Solve for u.

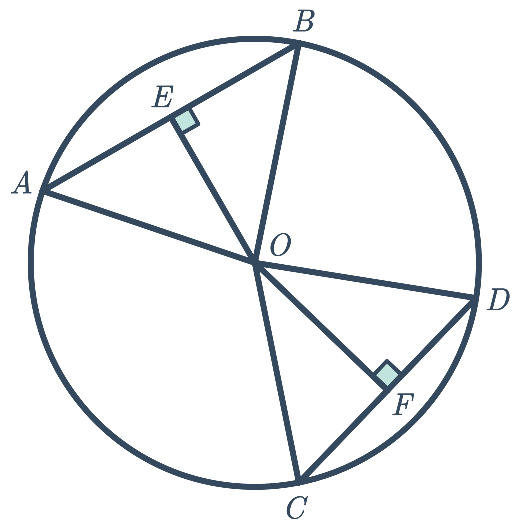


Proofs

12 Let O be the centre of the given circle with $AB = CD$.

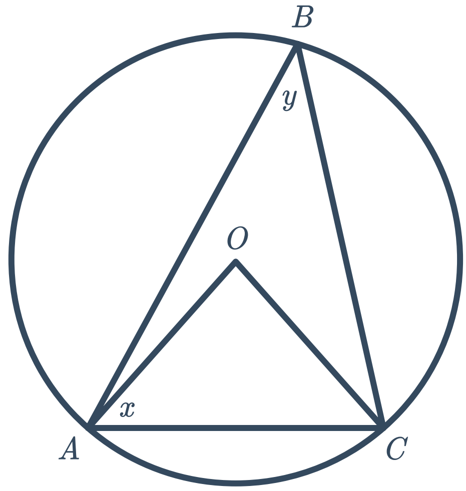
a Prove that $\triangle ABO$ and $\triangle CDO$ are congruent.

b Hence, prove that $\triangle BEO$ and $\triangle DFO$ are congruent.



13 (Extended)

Let O be the centre of the given circle:



Prove that x and y are complementary angles.